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Step 1: Register at github	If you don't have an account yet, first register at www.github.com <u>While registering use the same email that you configured in Git.</u> Check your configured email in GitBash terminal using command \$ git config user.email
Step 2: Configuring SSH Keys	You need to be authenticated on Github to do certain operations, like pushing up code from your local machine. Your terminal will prompt you every single time for your Github email and password, unless... You generate and configure an SSH key! Once configured, you can connect to Github without having to supply your username/password.
	Before generating SSH key check if you have an existing key. 1. Open Git Bash 2. Enter <u>ls -al ~/.ssh</u> to see if existing SSH keys are present. 3. Check the directory listing to see if you already have a public SSH key. By default, the filenames of supported public keys for GitHub are one of the following. id_rsa.pub id_ecdsa.pub id_ed25519.pub Tip: If you receive an error that ~/.ssh doesn't exist, you do not have an existing SSH key pair in the default location. You can create a new SSH key pair in the next step. For more detail refer: Checking for existing SSH keys - GitHub Docs
	If you don't have a supported public and private key pair, or don't wish to use any that are available, generate a new SSH key. For more detail refer: Generating a new SSH key and adding it to the ssh-agent - GitHub Docs 1. Open Git Bash 2. Paste the text below, replacing the email used in the example with your GitHub email address. This creates a new SSH key, using the provided email as a label. <u>Eval "\$ssh-agent -s)" <<your_email@example.com??</u> When you're prompted to "Enter a file in which to save the key", you can press Enter to accept the default file location. Please note that if you created SSH keys previously, ssh-keygen may ask you to rewrite another key, in which case we recommend creating a custom-named SSH key. To do so, type the default file location and replace id_ALGORITHM with your custom key name. > Enter file in which to save the key (/c/Users/YOU/.ssh/id_ALGORITHM):[Press enter] At the prompt, type a secure passphrase. For more information, see "Working with SSH key passphrases." Or just press enter. > Enter passphrase (empty for no passphrase): [Type a passphrase] > Enter same passphrase again: [Type passphrase again]
Step 3: Adding	Either old key or new generated key, add the key to the ssh-agent.

your SSH key to the ssh-agent	<pre>// # start the ssh-agent \$ eval "\$(ssh-agent -s)" \$ ssh-add ~/.ssh/id_ed25519</pre>
Step 4: Add the SSH public key to GitHub account.	After you generate an SSH key pair, you must add the public key to GitHub.com to enable SSH access for your account. You can access and write data in repositories on GitHub.com using SSH (Secure Shell Protocol). When you connect via SSH, you authenticate using a private key file on your local machine. For more information https://docs.github.com/en/authentication/connecting-to-github-with-ssh/adding-a-new-ssh-key-to-your-github-account 1. Copy the SSH public key to your clipboard. If your SSH public key file has a different name than the example code, modify the filename to match your current setup. When copying your key, don't add any newlines or whitespace. # command copies the contents of the id_ed25519.pub file to your clipboard <pre>\$ clip < ~/.ssh/id_ed25519.pub OR \$ cat ~/.ssh/id_ed25519.pub clip</pre> 2. In the upper-right corner of any page, click your profile photo, then click Settings. 3. In the "Access" section of the sidebar, click "SSH and GPG keys" 4. Click "New SSH key" or "Add SSH key". 5. In the "Title" field, add a descriptive label for the new key. For example, if you're using a personal laptop, you might call this key "Personal laptop". 6. Select the type of key, either authentication or signing. 7. In the "Key" field, paste your public key 8. Click Add SSH key. 9. If prompted, confirm access to your account on GitHub.
Step 5: Test connection	<pre>\$ ssh -T git@github.com</pre>

```
prash@LAPTOP-A7SH541A MINGW64 ~
$ ssh-keygen -t ed25519 -C [redacted]@gmail.com
Generating public/private ed25519 key pair.
Enter file in which to save the key (/c:/Users/prash/.ssh/id_ed25519):
Created directory '/c:/Users/prash/.ssh'.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /c:/Users/prash/.ssh/id_ed25519
Your public key has been saved in /c:/Users/prash/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:2AZKZ6vuNkriAghw913ScD5xvneAaxW4H/WTQxkRX7s [redacted]@gmail.com
The key's randomart image is:
+--[ED25519 256]--+
|      o..o...+*|
|      ..+o++..++|
|      o=o..o..o+|
|...+ *o.....+o|
|+   .o S. ....Eo|
|o      . . . . .|
|oo.o          |
|...+o.         |
+---[SHA256]-----+

prash@LAPTOP-A7SH541A MINGW64 ~
$ eval "$(ssh-agent -s)"
Agent pid 2059

prash@LAPTOP-A7SH541A MINGW64 ~
$ ssh-add ~/.ssh/id_ed25519
Identity added: /c:/Users/prash/.ssh/id_ed25519 ([redacted]@gmail.com)

prash@LAPTOP-A7SH541A MINGW64 ~
$ cat < ~/.ssh/id_ed25519.pub | clip

prash@LAPTOP-A7SH541A MINGW64 ~
$ ssh -T git@github.com
The authenticity of host 'github.com (140.82.114.3)' can't be established.
ED25519 key fingerprint is SHA256:+DiY3wvvV6TuJJhbpZisF/zLDA0zPMSvHdkr4UvCoQu.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? Y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
Hi [redacted]! You've successfully authenticated, but GitHub does not provide shell access.

prash@LAPTOP-A7SH541A MINGW64 ~
$
```

